Data Structures and Algorithms

TP03: Pointer

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01

tp3_ex1.cpp

Week03 > lab03 > tp3_ex1.cpp > ...

You, 14 hours ago | 1 author (You)

```cpp
#include <iostream>
using namespace std;

int main() {

    int n1 = 7;
    int n2 = 3;
    int n3 = 15;

    int *p1 = &n1;
    int *p2 = &n2;
    int *p3 = &n3;

    // Part a: Display address and value of n1, n2, n3 using pointers
    cout << "Part a: Display address and value using pointers" << endl;
    cout << "n1 - Address: " << p1 << ", Value: " << *p1 << endl;
    cout << "n2 - Address: " << p2 << ", Value: " << *p2 << endl;
    cout << "n3 - Address: " << p3 << ", Value: " << *p3 << endl;

    cout << endl;

    // Part b: Update n3 using p3 to be the sum of n1 and n2
    cout << "Part b: Update n3 using p3" << endl;
    cout << "Before: n3 = " << n3 << endl;
    *p3 = *p1 + *p2;  // n3 = n1 + n2 using pointers
    cout << "After: n3 = " << n3 << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdap
cher.exe' '--stdin=Microsoft-MIEngine-In-p0kkycoy.nn5' '--stdout=Microsoft-MIEngine-Out-dyw34rcs.s4p' '--stderr=Microsoft-MIEngine-Er
Microsoft-MIEngine-Pid-12lx31sx.v2n' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Part a: Display address and value using pointers
n1 - Address: 0x5ffe64, Value: 7
n2 - Address: 0x5ffe60, Value: 3
n3 - Address: 0x5ffe5c, Value: 15

Part b: Update n3 using p3
Before: n3 = 15
After: n3 = 10
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

02

tp3_ex2.cpp

Week03 > lab03 > tp3_ex2.cpp > ...

You, 14 hours ago | 1 author (You)

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter a number: ";
    cin >> n;

    // Create pointer to n
    int *p = &n;

    cout << "Original value of n: " << n << endl;

    // Modify n to n+7 using pointer operation
    *p = *p + 7;

    // Display the modified value
    cout << "Modified value of n: " << n << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdap
--stdin=Microsoft-MIEngine-In-txc01i24.er2' '--stdout=Microsoft-MIEngine-Out-uth4vkhb.wtu' '--stderr=Microsoft-MIEngine-Error-mee5n0g
-ql0qxb0m.a3q' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter a number: 12
Original value of n: 12
Modified value of n: 19
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

03

tp3_ex3.cpp

Week03 > lab03 > tp3_ex3.cpp > ...

You, 14 hours ago | 1 author (You)

```cpp
#include <iostream>
using namespace std;

void exchange(float *a, float *b) {

    float temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    float num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "\nBefore exchange:" << endl;
    cout << "num1 = " << num1 << ", num2 = " << num2 << endl;

    // Call exchange function with pointers
    exchange(&num1, &num2);

    cout << "\nAfter exchange:" << endl;
    cout << "num1 = " << num1 << ", num2 = " << num2 << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm> & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAda
--stdin=Microsoft-MIEngine-In-u4wd3lye.4dk' '--stdout=Microsoft-MIEngine-Out-5p5rye0c.tem' '--stderr=Microsoft-MIEngine-Error-sfo1jo
-42onnxmo.fjs' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter first number: 35
Enter second number: 23

Before exchange:
num1 = 35, num2 = 23

After exchange:
num1 = 23, num2 = 35
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

04

tp3_ex4.cpp

Week03 > lab03 > tp3_ex4.cpp > main()

You, 14 hours ago | 1 author (You)

```cpp
#include <iostream>
#include <cmath>
using namespace std;

void solveEquation(int a, int b, int c, float *x1, float *x2, float *delta) {

    *delta = b * b - 4 * a * c;

    if (*delta >= 0) {
        *x1 = (-b + sqrt(*delta)) / (2 * a);
        *x2 = (-b - sqrt(*delta)) / (2 * a);
    } else {
        *x1 = 0;
        *x2 = 0;
    }
}

int main() {
    int a, b, c;
    float x1, x2, delta;

    cout << "Solve quadratic equation: ax^2 + bx + c = 0" << endl;
    cout << "Enter coefficient a (a != 0): ";
    cin >> a;

    if (a == 0) {
        cout << "Error: a cannot be 0 for quadratic equation!" << endl;
        return 1;
    }

    cout << "Enter coefficient b: ";
    cin >> b;

    cout << "Enter coefficient c: ";
    cin >> c;

    // Solve the equation using function with pointers
    solveEquation(a, b, c, &x1, &x2, &delta);
```

```cpp
    // Solve the equation using function with pointers
    solveEquation(a, b, c, &x1, &x2, &delta);

    cout << "\nResults:" << endl;
    cout << "Delta (Discriminant): " << delta << endl;

    if (delta < 0) {
        cout << "No real roots (discriminant is negative)" << endl;
    } else if (delta == 0) {
        cout << "One root: x = " << x1 << endl;
    } else {
        cout << "Two roots:" << endl;
        cout << "x1 = " << x1 << endl;
        cout << "x2 = " << x2 << endl;
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdapters\b
Solve quadratic equation: ax^2 + bx + c = 0
Enter coefficient a (a != 0): 1
Enter coefficient b: 5
Enter coefficient c: 3

Results:
Delta (Discriminant): 13
Two roots:
x1 = -0.697224
x2 = -4.30278
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

05

tp3_ex5.cpp M

Week03 > lab03 > tp3_ex5.cpp > main()

```cpp
#include <iostream>
using namespace std;

void findMaxMin(int* arr, int *max, int *min) {

    *max = arr[0];
    *min = arr[0];

    for (int i = 1; i < 7; i++) {
        if (arr[i] > *max) {
            *max = arr[i];
        }
        if (arr[i] < *min) {
            *min = arr[i];
        }
    }
}
int main() {
    int numbers[7];
    int maxValue, minValue;

    cout << "Enter 7 integer numbers:" << endl;
    for (int i = 0; i < 7; i++) {
        cout << "Number " << (i + 1) << ": ";
        cin >> numbers[i];
    }

    // Find max and min values using function with pointers
    findMaxMin(numbers, &maxValue, &minValue);

    cout << "\nArray contents: ";
    for (int i = 0; i < 7; i++) {
        cout << numbers[i] << " ";
    }
    cout << endl;
    cout << "\nMaximum value: " << maxValue << endl;
    cout << "Minimum value: " << minValue << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdap
--stdin=Microsoft-MIEngine-In-5ekrtshm.etk' '--stdout=Microsoft-MIEngine-Out-dtptyfn1.3tl' '--stderr=Microsoft-MIEngine-Error-kb35ehq
-my3wjjqt.qhb' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Array contents: 12 13 14 53 36 45 56

Maximum value: 56
Minimum value: 12
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

06

tp3_ex6.cpp

Week03 > lab03 > tp3_ex6.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main() {
    float numbers[7];
    float *ptr = numbers;

    cout << "Enter 7 float numbers:" << endl;
    for (int i = 0; i < 7; i++) {
        cout << "Number " << (i + 1) << ": ";
        cin >> *(ptr + i);
    }

    cout << endl;

    // Part a: Display all numbers using pointer variable
    cout << "Part a: Display all numbers using pointer operation:" << endl;
    float *displayPtr = numbers;
    for (int i = 0; i < 7; i++) {
        cout << *(displayPtr + i) << " ";
    }
    cout << endl << endl;

    // Part b: Compute summation and multiplication
    cout << "Part b: Compute summation and multiplication:" << endl;

    float sum = 0;
    float product = 1;

    for (int i = 0; i < 7; i++) {
        sum += *(ptr + i);
        product *= *(ptr + i);
    }

    cout << "Summation of all numbers: " << sum << endl;
    cout << "Multiplication of all numbers: " << product << endl;

    return 0;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAda
--stdin=Microsoft-MIEngine-In-lhg3caiz.c45' '--stdout=Microsoft-MIEngine-Out-yajwmult.kud' '--stderr=Microsoft-MIEngine-Error-ar1frr
-5nu5keb3.3qe' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Part a: Display all numbers using pointer operation:
12.5 15 32.2 54 21 5.32 6.3

Part b: Compute summation and multiplication:
Summation of all numbers: 146.32
Multiplication of all numbers: 2.29468e+08
```

07

tp3_ex7.cpp

Week03 > lab03 > tp3_ex7.cpp > ...

You, 14 hours ago | 1 author (You)

```cpp
#include <iostream>
using namespace std;

void sum1(double *sum, int n) {
    // Calculate sum of 1/1^2 + 1/2^2 + 1/3^2 + ... + 1/n^2
    *sum = 0;

    for (int i = 1; i <= n; i++) {
        *sum += 1.0 / (i * i);
    }
}

int main() {
    int n;
    double result;

    cout << "Enter a positive integer n: ";
    cin >> n;

    if (n <= 0) {
        cout << "Error: n must be a positive integer!" << endl;
        return 1;
    }

    // Call sum1 function to calculate the sum
    sum1(&result, n);

    cout << "Sum of 1/1^2 + 1/2^2 + 1/3^2 + ... + 1/" << n << "^2 = " << result << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdap
--stdin=Microsoft-MIEngine-In-ihng2d4a.0kp' '--stdout=Microsoft-MIEngine-Out-emkjab2q.war' '--stderr=Microsoft-MIEngine-Error-ppfv532
-qvkbqfe0.pwa' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter a positive integer n: 12
Sum of 1/1^2 + 1/2^2 + 1/3^2 + ... + 1/12^2 = 1.56498
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

Github Link:

https://github.com/mengly-eg/data-structures-and-algorithm.git